

Vedant Mane

vedant12mane@gmail.com | (857)-565-5980 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Data Engineer and AI Developer with 3+ years architecting production platforms using SQL, Python, Databricks, and Snowflake. Specializes in real-time ETL pipelines, dimensional data warehouses (Star Schema, SCD Type 2, bridge tables), and multi-agent AI systems with LangGraph and RAG architectures. Proven expertise in Oracle GoldenGate CDC, Apache Airflow, dbt transformations, and data quality frameworks.

EDUCATION

Northeastern University, Master of Science in Information Systems (GPA: 3.88/4)

Aug 2024 - May 2026

- **Relevant Coursework:** Data Management & Database Design, Designing Advanced Data Architecture for Business Intelligence, Data Science Engineering, Big Data Systems and Intelligence Analytics, Prompt Engineering & Generative AI, Business Process Engineering

University of Mumbai, Master of Science in Computer Science (GPA: 9.56/10)

Jun 2019 - May 2021

University of Mumbai, Bachelor of Science in Computer Science (GPA: 8.77/10)

Jun 2016 - May 2019

SKILLS

Programming Languages & Database: Python, R, Javascript, SQL, PL/SQL, Oracle, PostgreSQL, Pinecone, ChromaDB

Machine Learning & AI Frameworks: PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, Scikit-learn, Scipy, Claude Code

Big Data Engineering: Airflow, Spark, Databricks, Snowflake, dbt, ETL/ELT, CDC, Data Warehousing, Star Schema, Data Quality

Cloud Platforms & DevOps: AWS (S3, EC2), Google Cloud Run, Compute Engine, Microsoft Azure, Docker, Kubernetes, Git, CI/CD

Data Analysis & Visualization: Tableau, Power BI, Streamlit, FastAPI, Matplotlib, Seaborn, Navicat Data Modeler

EXPERIENCE

Graduate Teaching Assistant, Northeastern University | Boston, MA

Sept 2025 - Apr 2026

- Mentored 80+ graduate students across Data Management & Database Design and Prompt Engineering & Generative AI, delivering hands-on instruction in Oracle SQL, PL/SQL, ER and dimensional modeling, RAG architecture, and prompt engineering patterns.
- Evaluated 30+ ER diagrams, 250+ database assignments, and 100+ AI application projects, giving structured feedback that helped students build production-ready LLM applications.

Data Engineer, Tata Consultancy Services | Mumbai, India

May 2021 - July 2024

- Optimized Oracle GoldenGate CDC replication and warehouse transformations by refactoring 50+ SQL queries with index tuning, improving query performance by 20% and reducing data latency by 12%.
- Automated archival of high-volume transaction tables with Bash scripts scheduled through Autosys, referencing archived data via Oracle database links to keep active tables lean and sustain pipeline performance across a platform processing 450M+ daily transactions.
- Engineered production monitoring across 12 servers using Python and Bash for GoldenGate trail file parsing, anomaly detection, and alerting on transaction delays, stuck transactions, and service health, improving operational efficiency by 15%.
- Established Git-based CI/CD with automated testing for PL/SQL deployments, reducing release cycles by 30% and improving reliability through proactive detection of performance bottlenecks.

ML & AI Software Engineer Intern, ABCOM Information Systems | Mumbai, India

Feb 2021 - May 2021

- Built automated Machine Learning and Deep Learning workflows for model selection, hyperparameter tuning, and deployment across regression and classification tasks; reduced deployment time and improved experimentation cycles through data pipeline automation.

PROJECTS

Multi-Agent AI System for Startup Intelligence (LangGraph, Airflow, MCP, RAG, AWS, OpenAI, Pinecone)

[GitHub](#)

- Architected multi-agent orchestration using LangGraph state machines coordinating specialized agents (Market Analyst, Competitor Research, Location Intelligence, Financial Advisor) powered by OpenAI GPT-4o-mini.
- Implemented location intelligence using Model Context Protocol (MCP) server for Google Maps integration; developed scoring algorithm analyzing demographic data, competitor density, state regulations, and investment trends to generate data-driven site recommendations.
- Unified unstructured and structured data from Snowflake via Airflow pipelines into Vector DB, enabling contextual AI reasoning and semantic Q&A natural language processing; deployed FastAPI backend using a docker image on GCP Cloud Run with GitHub Actions.

SEC Financial Data Engineering (Airflow, dbt Cloud, Snowflake, AWS, GCP, FastAPI, Streamlit)

[GitHub](#)

- Built end-to-end financial data pipeline automating SEC EDGAR quarterly filing extraction (10-Q, 10-K) for 25 companies; architected Apache Airflow DAGs processing 50M+ financial values through CSV, JSON, and DBT transformations with parallel task execution.
- Designed data transformation layer with 7 models creating dimensional fact tables from raw SEC XBRL data; implemented data quality validation, schema tests, and business rule checks achieving zero data quality failures across 30M+ rows spanning 20 quarters.
- Built full-stack financial analytics platform with containerized FastAPI backend, Streamlit UI, AWS S3 integration, and Snowflake data warehouse; optimized costs achieving 81% reduction through warehouse auto-suspend, query efficiency, and storage tiering strategies.

Food Inspections Data Engineering & Analytics (Databricks, PySpark, SQL, Tableau, Alteryx, Navicat)

[GitHub](#)

- Engineered end-to-end ETL pipeline using Alteryx and Databricks Delta Live Tables with Change Data Capture (CDC) to unify 450K food inspection records from Chicago and Dallas with incompatible schemas into analytical data warehouse using Medallion Architecture.
- Performed data modeling to design Star Schema with 7 dimensions, fact table, and bridge table; applied SCD Type 2 to track historical changes; implemented data quality checks validating 97.8% of records had complete required fields and standardized formats.
- Developed interactive Tableau dashboard visualizing inspection pass/failure rates, violation trends, and geographic clustering of high-risk establishments; identified seasonal patterns in foodborne illness and correlation between facility size and inspection outcomes.

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Financial PDF Document Processing (Airflow, RAG, Mistral, Pinecone)

[GitHub](#)

- Engineered Retrieval Augmented Generation system processing NVIDIA quarterly reports through Airflow pipeline; implemented PDF extraction with Mistral OCR, AWS S3 storage, and OpenAI embedding generation for 8,307 vectors across 1,847 pages.
- Designed and compared three chunking strategies (Markdown-based sectioning, semantic chunking and sliding window) achieving 85-92% retrieval relevance; built Pinecone vector database enabling semantic search and metadata-based filtering.
- Developed financial intelligence platform with Streamlit UI enabling natural language querying of quarterly reports; revealed insights through temporal analysis including revenue growth, gross margin expansion, and AI business transformation visible from 2020 to 2024.

FRED Economic Data Analysis (Snowflake Tasks, Snowpark, SnowSQL, GitHub Actions, AWS)

[GitHub](#)

- Built automated economic data pipeline processing 10 years of Federal Reserve Treasury yield data using Snowflake-native orchestration; designed three-layer architecture (RAW → HARMONIZED → ANALYTICS) with Snowflake Tasks DAG, Snowpark-based notebooks.
- Implemented incremental data processing using Snowflake Streams for Change Data Capture and MERGE operations via stored procedures; created UDFs calculating yield spread and aggregated tables for recession indicator analysis and economic trend monitoring.
- Developed CI/CD pipeline with GitHub Actions scraping FRED API daily, storing data in AWS S3, and triggering Snowflake Tasks orchestration; built Streamlit visualization dashboard with unit tests validating UDF accuracy and stored procedure correctness.

IMDb Entertainment Analytics (Databricks, DQX, Delta Live Tables, Tableau, Lakehouse)

[GitHub](#)

- Engineered end-to-end streaming ETL pipeline using Databricks Delta Live Tables processing 98M+ records across 24 IMDb tables; implemented Medallion Architecture with PySpark transformations, data quality profiling via DQX, and automated schema evolution.
- Designed Star Schema with 12 dimensions, 3 fact tables, and bridge tables managing many-to-many relationships including 21M+ person-profession associations; modeled temporal hierarchies and slowly changing dimensions using Navicat Data Modeler.
- Built interactive Tableau dashboards analyzing 9M titles across 230 regions and 91 languages, revealing content distribution patterns, genre evolution trends, regional release strategies, and rating correlations for entertainment industry insights.

LA Crime Analytics Platform (Databricks, Delta Live Tables, Unity Catalog, Tableau)

[GitHub](#)

- Engineered end-to-end streaming pipeline using Databricks Delta Live Tables with Auto Loader to process 1M+ LAPD crime incidents (2020-2025) through Medallion Architecture with Unity Catalog for metadata management and data lineage tracking.
- Designed Star Schema with 8 dimensions and 1M-row fact table across 21 LAPD areas; implemented DLT Expectations framework in Silver layer achieving 100% data retention and quality validation across all pipeline stages.
- Built 6 interactive Tableau dashboards analyzing crime trends, geographic hotspots, temporal patterns, weapon usage, demographics, and arrest performance; total pipeline execution completed in 1 minute 48 seconds on Databricks serverless compute.

Academic Research Assistant (CrewAI, OpenAI, SerperDev, FastAPI, Streamlit)

[GitHub](#)

- Built multi-agent research assistant using CrewAI orchestrating 4 agents in sequential workflow; implemented custom Academic Source Analyzer tool with credibility scoring, bias detection, citation generation and automated literature review capabilities
- Designed agent coordination architecture with shared memory system enabling inter-agent communication and context persistence; integrated tools (web search, website scraping, file processing) and deployed APIs with Streamlit for natural language research queries
- Automated academic research workflow processing queries through planning, information gathering, analysis, and synthesis phases; generated structured reports with source evaluation, pattern identification, and comprehensive citations in standardized formats

LoRA/PEFT Fine-Tuning for Conversational AI (PyTorch, HuggingFace, LLM, LoRA, PEFT)

[GitHub](#)

- Fine-tuned 345M-parameter LLM for domain-specific mental health Q&A using Low Rank Adaptation (LoRA), reducing trainable parameters to 9.2M (2.67%) while maintaining response quality and cutting training time to 20 minutes.
- Implemented parameter-efficient fine-tuning (PEFT) pipeline with Hugging Face Transformers and PEFT library, reducing GPU memory requirements by 90% through 8-bit quantization and mixed precision training.
- Built safety-aware inference system with crisis keyword detection, automatic helpline routing, and ethical guardrails for Responsible AI deployment in sensitive healthcare domains.